PROJECT-2

AIM:- Configuration of route redistribution between RIP and EIGRP routing protocols using Cisco Packet Tracer.

◉ **STEPS :**

- **Router1 with RIP**
- **Router2 with EIGRP**
- **Router0 as the Redistribution Router** (connecting RIP and EIGRP)

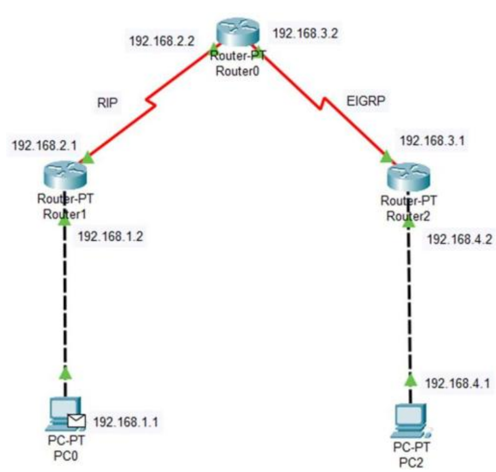
1. Assign IP Addresses to Each Device

On PC0 (Connected to Router1)

4. Open **PC0** in Packet Tracer.
5. Go to **Desktop > IP Configuration**.
6. Assign:
    - **IP Address:** `192.168.1.1`
    - **Subnet Mask:** `255.255.255.0`
    - **Default Gateway:** `192.168.1.2`

On PC1 (Connected to Router2)

4. Open **PC1** in Packet Tracer.
5. Go to **Desktop > IP Configuration**.
6. Assign:
    - **IP Address:** `192.168.4.1`
    - **Subnet Mask:** `255.255.255.0`
    - **Default Gateway:** `192.168.4.2`



2. Configure Router Interfaces

Each router has multiple interfaces connecting to different networks. Configure them with the correct IP addresses.

On Router1

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.1.2  255.255.255.0
exit
interface Serial0/0/0
ip address 192.168.2.1  255.255.255.0
exit
```

On Router0

```
enable
configure terminal
interface Serial0/0/0
ip address 192.168.2.2  255.255.255.0
exit
interface Serial0/0/1
ip address 192.168.3.2  255.255.255.0
exit
```

On Router2

```
enable
configure terminal
interface FastEthernet0/0
ip address 192.168.4.2 255.255.255.0
no shutdown
exit
interface Serial0/0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

3. Configure RIP on Router1

Since **Router1** is using **RIP**, enable it and advertise the connected networks.

```
enable
configure terminal
router rip
version 2
network 192.168.1.0
network 192.168.2.0
no auto-summary
exit
```

4. Configure EIGRP on Router2

EIGRP uses Autonomous System (AS) numbers, so you must define one when configuring the protocol.

```
enable
configure terminal
router eigrp 100
 network 192.168.3.0 0.0.0.255
 network 192.168.4.0 0.0.0.255
 no auto-summary
exit
```

5. Configure Redistribution on Router0

To allow communication between RIP and EIGRP, configure bidirectional redistribution.
Redistribute RIP into EIGRP

- RIP doesn't have metrics like EIGRP, so when redistributing RIP into EIGRP, a metric must be defined (bandwidth, delay, reliability, load, MTU).
- Example metric values: 10000 100 255 1 1500

```
enable
configure terminal
router rip
version 2
network 192.168.2.0
no auto-summary
exit

router eigrp 100
 network 192.168.3.0 0.0.0.255
exit

router eigrp 100
 redistribute rip metric 10000 100 255 1 1500
exit

router rip
 redistribute eigrp 100 metric 2
exit
```

6. Verify Configuration

Check Routing Tables

Run the following commands to confirm routes are learned correctly:

On **Router1**:

```
show ip route rip
```

On **Router2**:

```
show ip route eigrp
```

On **Router0**:

```
show ip route
```

You should see both **RIP and EIGRP routes** in the table.

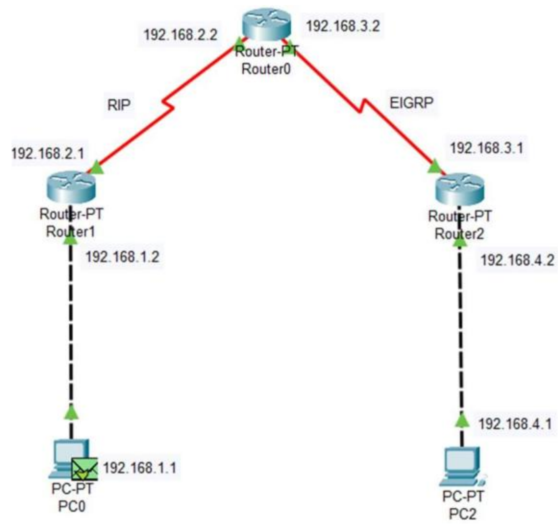
Check Neighbor Relationships

On **Router0**, verify EIGRP neighbors:

```
show ip eigrp neighbor
```

On **Router1**, verify RIP updates:

```
debug ip rip
```



7. Test Connectivity

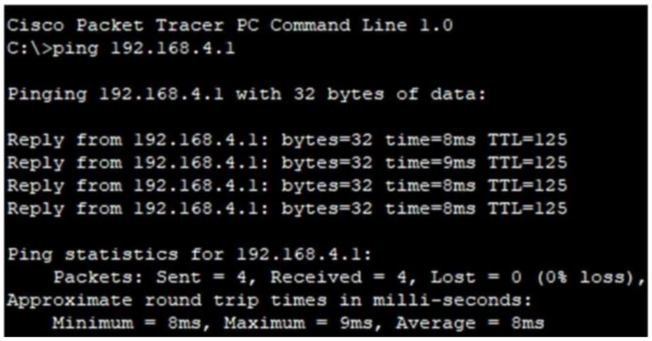
From **PC0 (192.168.1.1)**:

```
ping 192.168.4.1
```

From **PC1 (192.168.4.1)**:

```
ping 192.168.1.1
```

If both pings succeed, route redistribution is working corre



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.1

Pinging 192.168.4.1 with 32 bytes of data:

Reply from 192.168.4.1: bytes=32 time=8ms TTL=125
Reply from 192.168.4.1: bytes=32 time=9ms TTL=125
Reply from 192.168.4.1: bytes=32 time=8ms TTL=125
Reply from 192.168.4.1: bytes=32 time=8ms TTL=125

Ping statistics for 192.168.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 9ms, Average = 8ms
```